

Cueing and Cognitive Conflict in Judge–Advisor Decision Making

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The Judge–Advisor paradigm (Sniezek & Buckley, 1989) allows for the study of decision making by groups with differentiated roles. This paper reports a study using this approach to investigate the impact of advice vis-à-vis the judge's own initial choice. Teams consisting of one judge and two advisors (business students randomly assigned to these roles) were given a choice task (concerning business events) composed of 70 items with two alternatives each. The judges provided final team choices and confidence assessments under one of three conditions: Dependent (judge has no basis for own choice), Cued (judge chooses only after being advised), or Independent (judges makes own tentative choice prior to being advised as well as subsequent final choice). Results showed that this manipulation affected the judge's final choice accuracy and confidence, leading to the best performance by Independent judges and the poorest by Dependent judges. These data are discussed with respect to theory and data on the cueing effect (Sniezek, Paese, & Switzer, 1990) for individual choice. In addition, the effects on the decision making process of the judge from (a) advisors' confidence and (b) conflict between advisors' recommendations are examined in detail. Finally, issues concerning the potential contribution of the Judge–Advisor paradigm to the understanding of social decision making are addressed.

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There are many situations in which more than one person is involved in making a judgment or choice under conditions of uncertainty. The decision making of families, audit teams, city planners, personnel committees, physicians and patients, etc. are social by nature. In even the most personal of choices, one may take into

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account the opinions of others (Ajzen & Fishbein, 1980). However, when using current decision research and theory regarding accuracy and confidence to understand such social decision making, two concerns arise.

First, most research applies only to individual judgment or choice and, consequently, does not represent the social aspects of many important decision making situations. Second, the modest amount of literature on confidence and accuracy in social decision making is concerned almost exclusively with groups whose members' roles are undifferentiated (see Sniezek, 1992). But, role differentiation can be readily observed even in ad hoc experimental groups (Bales, 1950; Bales & Cohen, 1980). And, in social organizations, roles represent prescribed or standardized forms of activity (Katz & Kahn, 1966). Hence, group members often will begin a decision task in roles that regulate the timing, nature, and level of their participation in the decision process. Given this, it seems clear that many important decisions are made in a decision system that cannot be clearly dichotomized as either an individual or a group.

We seek to remedy this reliance on individual and group research, thereby avoiding some of the deficiencies associated with either extreme. To do this we combine the characteristics of each and we call the resulting structure the Judge–Advisor System (JAS).

In its most elemental form a JAS consists of one or more persons in the role of advisor who formulate judgments or recommend alternatives and communicate these to the person in the role of judge. A perfect example of a simple and common JAS structure is the reviewing process to which this paper will be subjected. An editor (the judge) chooses two reviewers (the advisors). The editor evaluates the comments and recommendations of the reviewers and makes the final judgment. The study reported here uses exactly that common structure—one judge and two advisors.

Although this paper is concerned with a specific common and simple JAS structure, it is useful to consider that, in its more complex forms, either role—advisor or judge—may be filled by a group or another JAS and many of these may be components of an intricate hier-

archy of social decision making. But, regardless of the particular form the JAS takes, it is the judge who has the responsibility for making the final decision and the decision has consequences for both the advisors and judge.

One major feature of Judge–Advisor research is that it allows us to consider the combined impact of factors primarily associated with only group or only individual research. Therefore, an important goal of Judge–Advisor research is to identify those variables associated with individual and group decision making that have the potential to enhance or degrade judgment or choice accuracy, to elevate or diminish confidence in those judgments and choices, or to influence the magnitude and direction of those judgments and choices in complex social decision systems.

Toward that end, this paper is concerned with the effects on accuracy and confidence of two variables. The first, *cueing*, is drawn from the individual decision making research and involves drawing a judge's attention to some alternative prior to choice (Ronis & Yates, 1987; Sniezek *et al.*, 1990). The second, *cognitive conflict*, is drawn from the groups research and signifies disagreement among individuals in their independent individual judgments (Rohrbaugh, 1979; Sniezek & Henry, 1989). In addition, this study allows us to answer questions regarding the effect of confidence on the choice process. The next section briefly reviews relevant studies on cueing and discusses their implications for JASs.

THE CUEING EFFECT

Ronis and Yates (1987) and Sniezek *et al.* (1990) arbitrarily or randomly preselected alternatives (i.e., provided an irrelevant cue) and directed subjects to provide subjective probabilities of correctness for the cued answers. Such cueing of one of the alternatives was found to raise overconfidence above the level observed in standard conditions, in which the subject independently selects the "correct" alternative. What is most striking is that in both of these studies, this "cueing effect" had opposite effects on accuracy and confidence—simultaneously making people less correct but more confident. To explain this finding, Sniezek *et al.* present a heuristic process model based on earlier theorizing by Koriati, Lichtenstein, and Fischhoff (1980). The essence of this model is that confidence and accuracy depend on the extent to which the decision maker searches for evidence about each alternative. Cueing is thought to make people less accurate but more confident about their accuracy because it reduces information processing about the uncued alternative. The resulting smaller samples of evidence will, in general,

reduce one's chance of selecting accurately. And they will also reduce the chance that the evidence considered is comprised of the contradictory pieces of information that would diminish confidence.

If the cueing effect generalizes to Judge–Advisor decision making, the judge's accuracy and confidence will vary as a result of the characteristics of the environment in which the decision is made. Some environments allow for and encourage more consideration of information about alternatives than other environments. To show how this can occur, we begin by distinguishing between two potential sources of information available to a judge, *internal* and *external*, and show various ways in which these sources can be combined and separated to create different environments for the consideration of alternatives.

The judge's own knowledge concerning the decision problem and alternatives constitutes an internal source of information while the advice of others is an external information source. A judge with advisors usually has the potential to access both sources of information. But the extent to which each is accessed and utilized by the judge in the decision process may depend on the timing of the advice. A judge who makes a preliminary choice and then receives an advisor's recommendation must use only the internal source for the initial choice, and has the option of using the external source in making the final choice. A judge who has first been cued by an advisor's recommendation and then makes a choice also has the opportunity to use information from both external and internal sources. Yet, the cued judge could conceivably use only the external information and forego an independent evaluation of the alternatives using internal information.¹ This is what the data on the cueing effect and model presented by Sniezek *et al.* (1990) suggest. And, as a consequence of the reduced consideration of information about alternatives, judges who have been cued are likely to be more confident—but not necessarily more accurate—than uncued peers.

To see how the cueing effect may operate in social decision making, the present study examines three distinct environments in which a judge chooses between two alternatives after receiving input from advisors. The three conditions representing different environments vary with respect to the source and timing of the information available to the judge.

One condition is designed to represent the common situation in which a judge, with knowledge relevant to the decision task, considers alternative answers and

¹ The reverse is plausible also. That is, a Cued judge could use only the internal information source and ignore external information from advisors. However, there is no basis for predicting this under the conditions used in this research.

(tentatively) chooses among them prior to receiving the opinions of advisors. This might well be the case of an editor who reviews a manuscript personally before receiving the opinions of the reviewers. Thus unique feature of this condition is the initial independent choice by the judge; hence, it is labeled *Independent*. The Independent condition provides the richest of the three environments because it encourages maximal processing of information by presenting the option of accessing information from both sources, in the order: Internal, External, Internal.

The second condition of Judge-Advisor decision making occurs when the judge obtains advice before having the chance to access internal knowledge. This is called the *Cued* condition. An example might be an editor who receives the opinions of the reviewers before personally considering the manuscript. As in the Independent condition, the environment provided in the Cued condition allows for the option of accessing information from both sources. But here the order is: External, Internal. The critical difference between the Independent and Cued conditions is that in the Independent condition the judge *must* access internal knowledge and make an independent choice prior to receiving advice. Thus, the Independent condition provides access to a potentially richer environment than does the Cued condition.

The third type of environment of interest in this study is one in which judges cannot (or will not) bring their own knowledge to bear on a choice problem. In reality, this often occurs when a naive judge has expert advisors. An example might be an editor who has little knowledge of a particular area and is thereby extremely dependent on the reviewers. The critical feature of this environment is that the only information available to the judge is the external information supplied by the advisors; this, the judge is completely dependent on the advisors. To mimic this situation, the *Dependent* condition is created to maximize the differentiation between the use of internal and external knowledge. This environment is constructed by preventing the judges from seeing the question-stems that the advisors use to make choices. Admittedly, this particular manipulation to achieve a Dependent condition is artificial in that judges in real decision situations will typically be informed of the question—even though the question can often be meaningless to the naive judge. Yet, the Dependent judges in this study face effectively the same inability to access internal information and evaluate choice alternatives for themselves as when they do not have knowledge in the problem domain. Because the Dependent condition provides an environment in which no internal information can be used, it represents the opposite extreme of the Inde-

pendent condition. As such, performance in this condition provides a meaningful reference point for the evaluation of performance in the Independent and Cued conditions.

COGNITIVE CONFLICT

Disagreement among advisors—even expert advisors—is prevalent if the decision problem is full of uncertainty, if the advisors lack sufficient information, or if the advisors have different agendas. Advisor disagreement is a characteristic of collective advice that is expected to affect final judge confidence and accuracy. In research on group judgment, cognitive conflict or disagreement, (i.e., heterogeneous judgments among group members) has been related to significant process gain. That is, Snizek and Henry (1989) and Rohrbaugh (1979) report similarity of initial individual judgments to be inversely related to final group judgment quality. Further, a high variance in individual judgments has been associated with the magnitude of the difference between average individual and group judgments (Libby, Trotman, & Zimmer, 1987). In a study of judgmental forecasting by Snizek (1990), there was little opportunity for cognitive conflict because all group members were provided with the same information—time series data. This study showed no evidence of process gain. It should be noted that each of these studies required subjects to estimate a quantity. The current study assumes that these findings will generalize to a choice task.

With two advisors making recommendations from a set of two alternatives, Advisor Conflict becomes a dichotomous variable, with the two levels being Conflict and No Conflict. These two levels of Advisor Conflict are likely to affect judge behavior differently, and possibly even interact with the Environment manipulation. All else being equal, Conflict is expected to benefit performance more than No Conflict in the Judge-Advisor form of social decision making.

But of course, all else is rarely equal. It is unlikely that the judges' performance on items on which their advisors disagree will be better than on those items for which their advisors agree. This is because advisors will agree more often on easier items than they will on more difficult items. Thus, we can expect the accuracy of *initial* Independent judge choices to be somewhat lower on items for which advisors disagree than on those for which they agree. To avoid confounding of item difficulty, the appropriate comparison between Conflict and No Conflict items is in terms of the magnitude of differences in the performance measures between them relative to the respective base rate for those items from individuals (e.g., Independent judges prior to receipt of advice) for those same items.

No Conflict

We begin by examining the differences between environments on those items for which the two advisors agree. The model of choice and confidence proposed by Sniezek *et al.* (1990) predicts that Cued judges will exhibit the cueing effect. That is, Cued judges will on the average be more confident and less accurate (and therefore more overconfident) than Independent judges. The loss of accuracy in the Cued condition results from the non-independence of the Judge's choice; the judge is expected to capitulate to the influence of agreeing advisors more often in the Cued than in the Independent condition. Thus, it is unlikely that the total breadth and depth of the three parties' knowledge will be brought to the task in the Cued condition. An analogous situation occurs in groups when members make a decision as a group, without having made independent individual choices prior to group interaction. Sniezek and Henry (1990) found superior group judgment accuracy if individual judgments preceded the group judgments.

Assuming that the amount of knowledge brought to bear on the problem will affect performance, we can expect Dependent judges to perform at the lowest level. That is, Dependent judges should have the highest levels of matching (agreement with the advisors), and therefore be no more accurate than their advisors. And, they are expected to show the lowest choice accuracy, but highest confidence because they will be able to do the least amount of information processing about alternatives. In summary, when advisors agree, the environment manipulation is expected to produce inverse effects on accuracy and confidence, with the smallest discrepancy between accuracy and confidence (or best calibration-in-the-large; see Yates, 1990) occurring in the Independent condition, and the largest discrepancy between accuracy and confidence (or poorest calibration-in-the-large) occurring in the Dependent condition.

Conflict

The Sniezek *et al.* (1990) model predicts that disagreement in the form of opposing recommendations of advisors will have a generally positive effect on performance because it encourages the consideration of all information available in the environment. Hence, Independent judges will be more accurate and less (over)-confident than individuals for items on which their advisors are in conflict. In the Cued condition, conflict should effectively eliminate the cueing effect by forcing judges to process evidence about multiple alternatives. Consequently, conflict between advisors is expected to improve performance in the Cued condition so that choice accuracy and confidence are at the level of the

Independent condition, and higher than the baseline for individuals.

The Dependent judges also have some potential to benefit from disagreement between advisors. In the absence of conflict, Dependent Judges are expected to follow the single recommendation of their advisors. Thus, they cannot become more or less accurate than their advisors were as individuals. However, under conditions of disagreement between advisors, these judges have the opportunity to improve over average advisor choice accuracy. For the improvement in accuracy to be realized, advisor confidence assessments must be valid and available. Then, judges need only use the heuristic of selecting the choice recommended by the more confident advisor to become more accurate than advisors are on average. If confidence assessments are available, valid, and used, even judges in the Dependent environment can exceed the performance levels of individuals.

Another hypothesized benefit of advisor conflict is to reduce the confidence of the judge. This is based on the observation that learning about disagreement among others reduces one's own initial confidence and increases information processing (Sniezek & Kolzow, 1994). Because overconfidence is widespread in difficult choice tasks (see research summary below), reductions in confidence will typically improve calibration. Thus we anticipate that choices made by judges under Conflict will show a reduction in confidence and therefore improvement in calibration over individual unadvised choice (Independent judges prior to advise)—regardless of environment.

SOCIAL PROCESSES AND CONFIDENCE IN CHOICE

Decision making by multiple parties—as for individuals—presumably has the primary goal of maximizing judgment and choice quality (subject to constraints, of course). But because objective evaluation of decision quality is difficult or impossible to achieve in a timely fashion, subjective evaluation of decision quality is an important part of decision making. Thus, research on confidence in judgment and choice is crucial for understanding the decision making of individuals (Sniezek & Buckley, 1993), groups (Sniezek, 1992), and JASs. Following the approach used by Sniezek and Henry (1989), confidence will be examined here from two perspectives: (a) as an outcome of the decision process and (b) as a factor that influences the decision process.

Confidence as an Outcome

There is a large (and growing) literature on confidence assessments that addresses the question of calibration for individuals. The research is typified by

data on calibration-in-the-large, the match between people's average level of confidence and their average level of accuracy. The general conclusion is that individuals are not well calibrated in the assessment of their judgments and choices in difficult tasks, but instead tend to be overconfident (cf. Seaver, von Winterfeldt, & Edwards, 1978; Lichtenstein, Fischhoff, & Phillips, 1982), though there are exceptions (e.g., for experts in some domains, Keren, 1987; for assessments of aggregated choices, Snizek *et al.*, 1990 or judgments, Snizek & Buckley, 1991; also see Yates, 1990). Similarly, recent studies of social judgment have found people to be overconfident in their predictions for others and themselves (Dunning, Griffin, Milojkovic, & Ross, 1990; Vallone, Griffin, Lin, & Ross, 1990).

Fewer data exist on confidence assessments made by groups or by individuals in a social decision-making context. (For a review of group confidence research, see Snizek, 1992.) Research has revealed substantial differences in confidence between groups as a whole and their individual members (e.g., Boje & Murnighan, 1982; Snizek & Henry, 1989, 1990; Snizek & Kolzow, 1994; Snizek, 1989). The most reliable finding from these studies is that social interaction increases group members' confidence. But because decision performance may be improved by social interaction as well, overconfidence can actually be reduced (as it was in Snizek & Henry, 1989). Similar to the approaches used in these studies, the present study will compare judges' choice accuracy and confidence after being advised to their pre-advice choice accuracy and confidence. In addition, between-subjects comparisons of judges' final confidence levels and choice accuracy will be made across various types of information processing environments.

Confidence as Influence

An assumption of this paradigm, based on Deutsch and Gerard (1955), is that confidence is a mechanism of influence between advisors and the judge during the decision-making process (Buckley & Snizek, 1990). Confidence enhances one's influence on others while uncertainty makes one susceptible to social influence. Thus, in social systems that make decisions under uncertainty, confidence is power.

Two questions about the role of advisor confidence in the decision process are important. The first is whether advisor confidence differs across variations of information processing environments. Because advisors are often aware of some details of the judge's choice situation, they may adjust their advice or confidence accordingly. To explore the possibility of such strategic action, advisor confidence will be treated as a dependent variable in analyses for the effects of the environ-

ment manipulation. Because we have no theoretical basis for differential predictions for the various conditions, no formal hypotheses about strategic action by advisors are offered.

The second important aspect of advisor confidence concerns its potential effect on the behavior of the primary actor in the system, the judge. If a advisor's confidence is communicated to the judge, it is expected to influence the judge's final choice. Judges who process more information are expected to be less persuaded by an advisor's confidence. Thus, we can expect that, when advisors make conflicting recommendations, the tendency of a judge to concur with the more confident advisor will be least in the Independent condition and greatest in the Dependent condition. Expressions of advisor confidence can assist the judge in making quality decisions only if two criteria are met: (a) *validity* (i.e., confidence assessments are systematically related to the quality of the advice) and (b) *utilization*, (i.e., the judge's use of the advice is related to the advisor's confidence level). The present research seeks to describe the influence of advisor confidence in the various conditions of Environment. In addition to tracking modifications in judge confidence as a function of advice, we shall identify the conditions that cause judges to change their initial choices.

SUMMARY AND HYPOTHESES

The goal of this research is to identify factors that affect the choices made, the accuracy of those choices, and the confidence placed in those choices within a Judge-Advisor System. Performance will be compared over conditions created by the manipulation of Environment (Independent, Cued, and Dependent) and Advisor Conflict (Conflict or No Conflict). If the match between choice quality and assessed choice quality (calibration-in-the-large) is critical, overconfidence is the dependent variable of interest and follows directly from the examination of accuracy and confidence. It is useful to learn if a reduction in overconfidence resulted from increased accuracy, decreased confidence, or some combination of changes in the two variables. Thus, all three—accuracy, confidence, and overconfidence—are important outcome variables in this study. Hypotheses are:

1. Judge's choice accuracy (i.e., % correct) will be higher when the information processing environment is richer. Therefore,
 - 1a. A main effect for Environment is expected such that Independent > Cued > Dependent.
 - 1b. With No Conflict, a simple main effect of Environment is expected such that Independent > Cued > Dependent.
 - 1c. With Conflict, a simple main effect of Environment is expected such that Independent = Cued > Dependent.

2. Relative to the baseline for individual performance, Conflict between advisors is expected to improve the judges' accuracy.
3. Judges' confidence (i.e., mean probability) will be higher when the information processing environment is poorer. Therefore,
 - 3a. A main effect for Environment is expected such that Dependent > Cued > Independent.
 - 3b. With No Conflict, a simple main effect of Environment is expected such that Dependent > Cued > Independent.
 - 3c. With Conflict, a simple main effect of Environment is expected such that Dependent > Cued = Independent
4. Relative to the baseline for individual performance, Conflict between advisors is expected to decrease the judges' confidence.
5. With No Conflict, judges will vary in the degree to which their choices match the advisors such that Dependent > Cued > Independent.
6. With Conflict, judges will vary in the degree to which their choices match those of the advisor with the higher level of confidence such that Dependent > Cued > Independent.

METHOD

Three experimental conditions of Environment (Independent, Cued, and Dependent) were designed to represent varying types of information processing situations. The second factor, Advisor Conflict (No Conflict, Conflict), was created naturally by the recommendations of the two advisors. A third factor, Judge Type, (experimental and control) was created by yoking an additional judge to each of the teams to examine judge behavior when advisor confidence is not communicated. Experimental judges received confidence assessments and the choices of their advisors, control judges received only the choices. The design was a $3 \times 2 \times 2$ (Environment $\times$ Advisor Conflict $\times$ Judge Type) factorial with Advisor Conflict a within-subjects factor.

Participants

Graduate and undergraduate students ($n = 131$) in hotel management, labor relations, and business programs were recruited for participation through advertisements and handouts explaining the general nature of the study and the reward system. To ensure that consequences of the judge's choices were important to all team members, there was an incentive for accuracy and it was described in the experimental as well as the recruitment materials. Participants whose overall team performance (in terms of choice accuracy) was in the top half of all teams in each of the six conditions were awarded money. The award levels were \$50 per person for first place and \$20 per person for second place. (Advisors for a winning judge in both the experimental and control conditions were not paid twice.) The remaining teams above the median for their conditions received \$10 per person.

Subjects were assigned to one of the three Environment conditions by the date and time of their partici-

pation. As subjects entered the laboratory, they were assigned to teams of three, and to the advisor and judge roles, through randomly ordered task materials. Each experimental condition included 11 groups of three members ($n = 99$). The number of control judges yoked to teams in the Independent, Cued, and Dependent conditions (10, 11, and 11, respectively) were unequal because one judge who failed to follow instructions was dropped. In addition, two control judges in the Dependent condition did not provide confidence assessments.

Procedure

There are many valid options for communication between judge and advisors, ranging from face-to-face discussion to anonymous electronic transmission of advice. Both extremes of social interaction have unique problems; variables extraneous to advice per se are numerous and roam freely in the former, possibly obscuring important effects. In the relatively brief relationship created in the laboratory, personal variables may be particularly problematic. In contrast, the asocial nature of electronic transmission reduces the chance that judges will appreciate that their advisors are in fact their peers from the subject population. We chose to begin our investigation of judge-advisor decision making with a moderate amount of social interaction in an effort to avoid some of the disadvantages of either extreme.

Participants completed the task in a room with others, but the identities of their team members were not revealed. This permitted us to control for variance based on perceptions of personal characteristics such as status or age, and to yoke the control judges to the experimental teams. Role assignments for all subjects in the room were identifiable from seating area and pen colors. All advisors were seated in the section nearest the door so that judges would notice their presence as they passed to their section at the far end of the room. Advisors were labelled "Brown" or "Blue" and given matching colored markers. Judges were referred to as "Judge Green" and given green markers. In the Independent condition, red markers were used for their initial responses and green markers for their final responses.

Upon arrival at the laboratory, each participant was given written instructions and a booklet containing task materials. General instructions for persons in all conditions gave details of the reward system, emphasizing that rewards would depend solely on the accuracy of final choices made by the team's judge, but would be distributed equally across team members. Those assigned to the role of judge in the Cued and Dependent conditions waited while their advisors com-

pleted the task. They were provided with magazines on travel to read as they waited (generally less than 20 min).

The written instructions also gave specific information about the task for each participant's role and experimental condition. Independent judges answered twice—before and after receiving their advisors' answer sheets. Cued and Dependent judges received their own task booklet simultaneously with the completed booklets of their advisors. Prior to marking their own choices, all judges were required to put a check mark next to the alternatives chosen by their advisors to insure that they had received these recommendations. In addition, Independent judges put a check mark next to their initial choices. Dependent judges were prevented from bringing their knowledge to the problem by withholding the item stems from them. Thus, Dependent judges saw only the two alternatives with advisor recommendations circled in blue and brown, and the accompanying confidence assessments. The procedure for the control judges in each condition was identical, except that they were not given the advisors' confidence assessments.

Task

The task consisted of a set of 70 two-alternative items that were selected by pretesting with volunteers (who had a chance in a \$20 lottery) drawn from the same population as were participants in the experiment (none participated in both the pretest and the experiment). Item difficulty levels ranged from .5 to .8 and averaged .6. Item content was based on information found in recent issues of *Business Week*, *The Wall Street Journal*, and *The New York Times*. (Sample items are: In the United States, 41% of all money spent on food is spent in (A) supermarkets or (B) restaurants? Which company has a larger share of the over-the-counter insecticide market? (A) Johnson Wax or (B) American Home Products.) Judges completed each item by circling (a) the letter of alternative chosen as correct and (b) a number on a probability scale ranging from .50 to 1.00 (in increments of .05) to indicate confidence in this choice.

To capture features inherent in actual Judge-Advisor decision situations, the task must have four characteristics: (a) the participants must have some, but not complete, relevant information so that they will have some uncertainty about the quality of their choices; (b) the participants must find the task to be interesting; (c) the task must produce outcome data that can be used to determine accuracy; and (d) there must be consequences of the quality of the judgment or choice task for all parties in the JAS. The construction of the task, the comments of the subjects, and the re-

ward system lead us to conclude that each criterion was met.

RESULTS

Judge accuracy, confidence, and overconfidence were dependent variables in a 3 (Environment: Dependent, Cued, Independent) $\times$ 2 (Judge Type: Experimental vs Control) $\times$ 2 (Advisor Conflict: No Conflict vs Conflict) ANOVA with level of Advisor Conflict a within-subjects factor. Following the details of these analyses, we report the effects of Advisor Conflict and confidence on final judge choices.

Before examining the accuracy scores of judges, it is necessary to know whether judges in the various experimental conditions of Environment received different advice. Analyses with advisor accuracy and confidence as dependent variables showed no main or interaction effects for Environment (all F 's less than one). Thus, we can proceed under the assumption that any observed differences in judge accuracy or confidence over the three conditions of Environment are not due to differences in advisor accuracy or confidence.

Judge Accuracy

In a choice task, accuracy corresponds to the percent of items answered correctly, sometimes called a knowledge score. Because analyses on judge accuracy and confidence showed no significant main or interaction effects due to Judge type, means reported below include combined data from Control and Experimental judges unless otherwise noted. However, further analyses were conducted to diagnose why the expressions of advisor confidence (available to experimental judges only) failed to make the experimental judges more accurate than the control judges.

The first potential cause, a lack of validity of advisor confidence assessments, was investigated by computing point biserial correlations between advisor choice accuracy (correct vs incorrect) and advisor confidence. Results show a positive but nonsignificant relationship in each of the three conditions of Environment, with an average $r = .15$ ($p > .05$) over all 66 advisors.² No differences in this relationship over conditions of Environment were found. Another plausible explanation for the failure of experimental judges to outperform control judges is that experimental judges ignored their advisors' confidence assessments. This possibility is explored later in the section labeled "Taking Advice."

² The pt. biserial correlation between accuracy and confidence was computed for each of the 66 advisors. These values were converted to Fisher's z scores, averaged, and converted back to the r scale for a mean r of .15. This process was used whenever correlations were averaged.

Condition means and significance tests for final judge accuracy and confidence are displayed in Table 1.

Environment. There was a significant main effect of Environment on judge accuracy and the interaction with Advisor Conflict was nonsignificant. Planned comparisons were used to test Hypothesis 1a concerning the effect of Environment. The difference in mean judge accuracy was significant for Independent vs Cued, Independent vs Dependent and approached significance for Cued vs Dependent. Thus, there was generally strong support for Hypothesis 1a.

With No Conflict between advisors, mean accuracy across conditions was in the anticipated direction; however, planned comparisons showed that only the difference between the Independent and Dependent conditions was significant. Thus, Hypothesis 1b was only modestly supported.

Under Conflict, the hypothesized superiority of the Independent judges over the Dependent judges was observed. But the Cued judges showed surprisingly poor performance. They were expected to perform at the level of the Independent judges so that judges in both of these conditions would surpass those in the Dependent condition. But instead, the Cued judges were only as accurate as the Dependent judges. As a result, both Cued and Dependent judges were inferior to the Independent judges. Thus Hypothesis 1c was not supported.

Advisor conflict. The main effect on accuracy for Advisor Conflict was significant ($F(1,62) = 28.4, p < .001$).

Collapsing across all three environments, judges exhibited a mean knowledge score of 66.6 when their advisors agreed and 58.4 when they disagreed.

However, recall that because advisors are more likely to be in agreement on relatively easier items, an a priori difference in item difficulty between the two levels of Advisor Conflict was anticipated. Such a difference in item difficulty indicates that the impact of advisor conflict is most appropriately assessed by comparing final judge accuracy levels to those of unaided individuals for items in the No Conflict and Conflict categories. Although technically they are not a control group, the accuracy of initial (pre-advice) Independent judges provides the best comparison with the final (post-advice) accuracy of judges. Initial Independent judges had access only to the same information as would any unadvised individual and they had no time limit. Accuracy under No Conflict and Conflict were not equal, but the difference was extremely small (60.1 with No Conflict and 59.8 with Conflict).

Final judge accuracy in each condition was then compared to the initial Independent judge accuracy. These planned comparisons demonstrated that, under No Conflict, advice was beneficial in all conditions (Independent: $t(20) = 5.43, p < .000$, Cued: $t(21) = 4.43, p < .000$, and Dependent: $t(21) = 2.09, p < .025$).

But, in contrast to Hypothesis 2, under Conflict, advice had only a small, nonsignificant, positive effect (Independent: $t(20) = .95, p < .18$) or a significant or marginally significant adverse effect on unaided accuracy (Cued: $t(21) = 1.98, p < .031$, and Dependent: $t(21) = 1.51, p < .073$).

Judge Confidence

Confidence for an individual subject is equal to the mean of that subject's confidence assessments over the set of items. Environment means for confidence are illustrated in Table 1. To make comparisons to the accuracy measure more straightforward, all confidence assessments were multiplied by 100.

Environment. Contrary to Hypothesis 3a, no main effect for Environment on confidence was found. But, the ANOVA revealed a significant interaction between Environment and Advisor Conflict ($F(2,60) = 13.32, p < .000$). The difference between Conflict and No Conflict confidence was least between the Independent conditions and greatest between the Dependent conditions (see Table 1).

The pattern of means provides some evidence in support of Hypothesis 3b in that less opportunity for processing information made judges more confident. With No Conflict, the Independent and Cued showed significantly lower levels of judge confidence than did the Dependent Condition. Though the direction of the dif-

TABLE 1
Accuracy and Confidence by Environment

	Environment			<i>F</i> (2,62)
	Independent	Cued	Dependent	
	Mean (<i>SD</i>)	Mean (<i>SD</i>)	Mean (<i>SD</i>)	
Accuracy				
Hypothesis 1a (Overall)	65.8 (5.4) ^{ab}	62.7 (5.4) ^a	60.6 (5.4) ^b	4.95**
Hypothesis 1b (No conflict)	69.1 (7.6) ^c	66.8 (7.1)	64.0 (8.8) ^c	2.31
Hypothesis 1c (Conflict)	61.7 (9.1) ^{de}	56.8 (7.2) ^d	56.8 (9.3) ^e	2.30
Confidence				
Hypothesis 3a (Overall)	73.1 (10.6)	73.6 (10.2)	74.7 (7.2)	.14
Hypothesis 3b (No Conflict)	75.0 (10.8) ^a	76.7 (11.8) ^b	82.6 (8.1) ^{ab}	3.02*
Hypothesis 3c (Conflict)	70.9 (10.8) ^c	69.1 (8.9)	64.8 (11.4) ^c	1.86

Note. Means with common superscripts differ from one another at $p < .05$.

* $p < .05$, ** $p < .01$.

ference in judge confidence between the Independent and Cued conditions was consistent with Hypothesis 3b, it was not significant.

When Conflict existed, the order of mean confidence across the three conditions of Environment was reversed (Independent > Cued > Dependent). The difference between Independent and Cued conditions was nonsignificant but the Independent was significantly different from the Dependent and the difference between the Cued and Dependent conditions approached significance. The direction of these differences were contrary to those predicted by Hypothesis 3c.

Advisor conflict. Using the same procedure as for accuracy, we calculated a baseline level of individual confidence for each level of Advisor Conflict from the mean initial Independent judge confidence (70.3 for No Conflict items and 68.5 for Conflict items). By comparing final judge confidence in each condition to this level of individual judge confidence, we can ascertain which manipulations produce more or less confidence in judges with advisors compared to unaided individuals. Mean judge confidence with No Conflict exceeded the baseline level of confidence (70.3) in all conditions. The difference was significant for all Environments (Independent: $t(20) = 2.00, p < .03$, Cued: $t(21) = 2.53, p < .01$, and Dependent: $t(19) = 6.81, p < .000$).

In contrast, the baseline individual confidence with Conflict (68.5) was not significantly different from mean individual confidence in any Environment. Thus, conflicting advice did not decrease confidence relative to the level of unaided individuals (contrary to Hypothesis 4), but it did prevent it from increasing.

Judge Overconfidence

The final performance measure is the discrepancy between confidence and accuracy. Although it is possible to observe directions of relative magnitude that demonstrate either over or underconfidence, the term overconfidence is used exclusively here because underconfidence was not observed.

The analysis on overconfidence revealed no significant main effect for Environment but a significant interaction between Environment and Advisor Conflict ($F(2,60) = 7.10, p < .002$).

Two findings underlie the interaction between Environment and Advisor Conflict on overconfidence. As seen in Fig. 1, the Dependent-No Conflict condition produced the highest level of overconfidence, although post hoc comparisons using Tukey's range test showed that it differed significantly only from Independent-No Conflict and Dependent-Conflict. But planned contrasts showed that, under No Conflict, overconfidence in the Independent ($t(60) = 3.63, p < .001$) and Cued

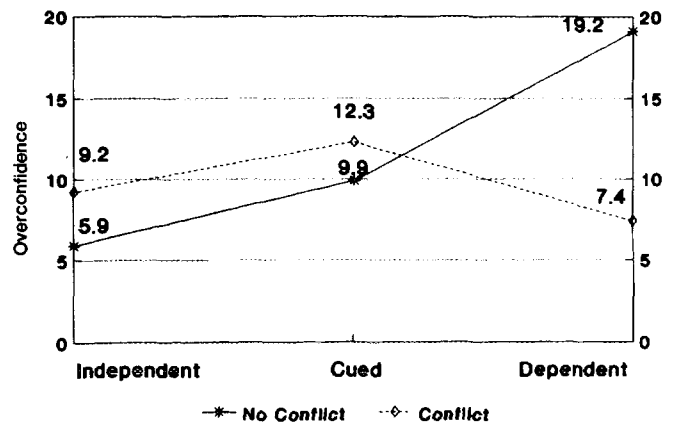


FIG. 1. Final judge overconfidence means as a function of Environment and Advisor Conflict.

($t(60) = 2.56, p < .007$) conditions was significantly lower than in the Dependent condition.

The Dependent condition produced the lowest level of overconfidence under Conflict, though this was not significantly different from other Conflict cells. Planned comparisons between No Conflict and Conflict for each Environment condition revealed significance only for the Dependent condition.

The second finding concerning the Environment $\times$ Advisor Conflict interaction concerns differences between the Cued and Independent conditions. A logical consequence of Hypotheses 1 and 2 is that Cued judges were expected to be more overconfidence than Independent judges under the No Conflict condition. Conflict, on the other hand, was expected to eliminate the cueing effect so that there would be no differences between the Independent and Cued Environments on accuracy or confidence, and hence, on overconfidence. Results show that overconfidence for Cued judges was greater than that of Independent judges, but not significantly greater (see Fig. 1), under either condition of Advisor Conflict.

Taking Advice

Hypotheses 5 and 6 were developed with the assumption that the extent to which judges take advice varies across conditions of Environment, with the strategy for using advice contingent on the level of Advisor Conflict. The proportions of judge choices consistent with each strategy are shown by experimental condition in Fig. 2.

Under No Conflict, judges were expected to employ the simple strategy of "matching" the choice recommended by the agreeing advisors. This strategy was anticipated to be most common in the Dependent condition and least common in the Independent condition. Planned comparisons (t tests on the mean proportion of matching by judges in different conditions of Environ-

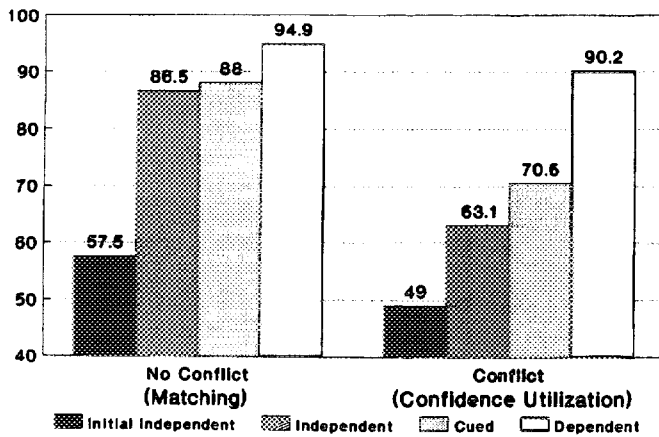


FIG. 2. Final judge choice strategies as a function of Environment and Advisor Conflict. Initial Independent (unadvised) judges are provided as a baseline for all conditions and to show the degree to which Independent judges changed their choices. Note: Matching strategy is for both Experimental and Control judges. Confidence Utilization strategy is for Experimental judges only. Control judges were not significantly different from chance (50%).

ment) indicated a significantly higher proportion of matching by Dependent (94.9%) judges than by either Cued (88%) ($t(62) = 2.42, p < .01$) or Independent (86.5) ($t(62) = 3.03, p < .002$) judges. Cued judges showed more matching than did Independent judges but the difference was not significant ($t(62) = .64, p < .27$). This tendency to match agreeing advisors was so strong that experimental judges whose advisors *both* gave confidence assessments of .5 (a guess) matched those advisors' choices 87% (47 of 54) of the time this scenario occurred.

Under Conflict, judges were expected to employ a "confidence utilization" strategy. This strategy required judges to select the alternative that was more confidently recommended when two advisors disagreed in their advice. Because control judges were not provided with the confidence assessments of their advisors, only data from the experimental judges could be used to test Hypothesis 6.

All three Environments exhibited levels of confidence utilization (i.e., a mean proportion of matches with the more confident advisor) in the expected direction. Independent ($M = 63.1$) differed from Cued ($M = 70.6$) ($t(30) = 1.63, p < .057$) and Dependent ($M = 90.2$) ($t(30) = 5.85, p < .000$) and Cued differed from Dependent ($t(30) = 4.22, p < .000$) providing substantial evidence in support of Hypothesis 6. Further tests demonstrated that these means were significantly different from .5 (chance) (for Independent $t(10) = 3.63, p < .03$.) As anticipated, means for control judges did not exceed chance in any condition. Thus, it is clear that experimental judges and control judges did not use the same process to make their choices. A more complete analysis follows in the next section.

Independent Judge Choice

No conflict. As can be inferred from Fig. 2, agreement between advisors often persuaded judges in the Independent condition to change their initial answers to agree with the advisors' answer. Judges initially agreed with advisors at no better than chance (57.5% of the time) but came to agreement on 86.5% of the items. However, advisor agreement affected the judges' confidence levels only slightly, increasing it from an initial level of 70.4 to 72.9, while accuracy increased from 60.1 to 69.1. This improved calibration-in-the-large subsequent to receiving advice.

Conflict. Independent judges initially agreed with the advisor who gave the high confidence assessment, again, only at the chance level (49%). Their final decisions, however, agreed with this advisor 63.1% of the time. The mean confidence of the advisor with high confidence was 79.6 while the judges' confidence increased from 66.9 to 69.7. Accuracy improved from 60 to 62 but as noted earlier this was nonsignificant. Control judges' mean confidence was 70.1 prior to receipt of the contradicting recommendations and 71.7 following it. Accuracy improved from only 60 to 61.

Cued Judge Choice

No conflict. In the Cued-No Conflict condition, judges followed suit 91.8% of the time (see Fig. 2). The judge was expected to show greater confidence than the advisors. This was true: judges' mean confidence was 76.1 while the advisors' was only 70.1.

Conflict. When advisors disagreed, their relative confidence was expected to have an influence on the judge. Judges chose the alternative with the higher confidence level 70.6% of the time. As expected, the judges displayed lower confidence in the chosen answer than did the advisor. Mean confidence for the advisors was 78.5 and mean confidence for the judges was only 71.7. When advisors disagreed and confidence assessments were unavailable, judges showed no preference for either advisor and expressed a mean confidence of only 68.

Dependent Judge Choice

No conflict. Under conditions of advisor agreement the judge chose that alternative in nearly all cases (94.9%) (see Fig. 2). As expected, judges exhibited higher confidence on these occasions than advisors by a wide margin. Judges' confidence was 82.6 and advisors only 67.9.

Conflict. When advisors disagreed and their confidence levels were available, the judge appeared to follow the confidence utilization strategy in 96% of the

cases. Just as in the Cued condition, the mean confidence level of the judges (68.1) was less than the mean confidence level of the advisors (77.3). As in the Cued condition, the control judges were random in the extent to which they matched their advisors. Control judges mean confidence was 63.1.

DISCUSSION

The first data we discuss are those related to the impact of advisor confidence assessments on judge choice. Next, factors affecting the three outcome variables—final judge choice accuracy, confidence, and overconfidence—are reviewed. Then we discuss the expression of uncertainty in the form of confidence assessments as social influence and strategy. We conclude with an evaluation of the Judge-Advisor design of the present study and the implications of this research for social decision-making theory and practice.

Given the lack of performance differences between experimental and control judges, we must conclude that the communication of uncertainty via confidence assessments by the advisors did not have a positive impact on the ultimate performance of judges. One apparent cause of the failure of confidence assessments to improve the performance of experimental judges is the strong tendency of judges to concur with advisors under conditions of No Conflict. On these occasions, advisors' confidence assessments had little effect on judges. Thus, under conditions of No conflict between advisors, experimental and control judges were similarly influenced by the recommendations of their advisors. Because over half (52%) of the observations involved agreement between advisors, a main effect of Judge Type would be difficult to achieve. Yet, the relatively greater impact of advisor consensus compared to advisor confidence is not a sufficient explanation for the lack of superior performance by experimental judges. There was no evidence of an interaction between Judge Type and Advisor Conflict; hence, confidence assessments yielded no statistically significant benefits even when advisors were in conflict.

Although judges who were given the confidence assessments of their advisors performed no better than those without them, there is strong evidence that the two types of judges behaved differently when there was disagreement between their advisors. Instead of ignoring available confidence assessments, judges in each condition of Environment were prone to accept the recommendation of the more confident advisor. Control judges had no such information to assist them in breaking the tie; thus, their process necessarily differed. So, the failure of confidence assessments to effect global performance enhancement cannot be due to the judges' reluctance to use them in the final choice pro-

cess. Instead, it was the low validity of these assessments that inhibited their value. The more confident advisor had only a modest chance of being more correct. Through low validities of confidence assessments have been reported (e.g., Sniezek, 1990; Sniezek & Henry, 1990), their consequences have not been addressed. If one of the purposes of expressing confidence is to communicate a subjective evaluation of the quality of a judgment or choice, then much can be gained by studying confidence assessments in social situations. This study gave all team members strong incentives to maximize final judge accuracy. Nevertheless, the advisors expressed confidence to their respective judges that was not only too high on average—but only very weakly related to the actual quality of the accompanying advice.

Judge Accuracy

Results show consistently, though not always significantly, that cueing had the predicted detrimental effect on performance. This occurred despite the Cued judges having had access to just as much information from both internal and external sources as did the Independent judges. The critical difference was in the order of accessing these sources. Independent judges had to consider both alternatives to provide their own initial choices. But, Cued judges could have considered only the advisors' recommendation. Results are consistent with those of Sniezek *et al.* (1990); manipulating the choice process to require greater processing of alternatives increased accuracy. Under No Conflict the accuracy means fell in the predicted order of Independent > Cued > Dependent. It was also anticipated that when advisors were in conflict, information processing would increase sufficiently enough to reverse the cueing effect so that Cued judges would exhibit accuracy at the same level as Independent judges. Contrary to our hypothesis, Cued judges who received conflicting advice were so impaired that they were no more accurate than the Dependent judges who were not permitted to access their own knowledge. It is as if cueing prevented the judges from accessing their own internal knowledge—even when it could have been used to resolve the conflicting recommendations of advisors.

One possible explanation for this is that the contradictory recommendations of the advisors presents a strong signal that the item is difficult. This may inhibit further information processing on the part of the Cued judge, who instead resorts to a choice heuristic for problems involving luck rather than skill. Potential heuristics—such as following the recommendation of the more confident advisor or guessing—are the same as those available to the Dependent judge. Hence, judge accuracy does not differ between the Cued and

Dependent conditions of Environment. The failure of disagreement to eliminate the cueing effect indicates that cueing may be even more detrimental to performance and more intractable than we previously stated. It appears that in this task judges did not, or could not, take advantage of their own relevant knowledge after being cued, even though there were large incentives for accuracy. Though teams in the Cued and Independent conditions brought equivalent amounts of knowledge to the task and enjoyed identical incentives for performance, the former were handicapped by the cueing of their judges.

Judge Confidence

Results for judge confidence show modest support for the Sniezek *et al.* (1990) model of choice and confidence in that Dependent judges had the least amount of information with which to make their final choices, yet they were the most confident under conditions of No Conflict between advisors. These results extend correlational research on expertise and probability assessment by showing that those with less knowledge can be more certain (Fischhoff & MacGregor, 1982) as well as more overconfident (Keren, 1987).

From one perspective, it is quite odd that judges who could not bring their own knowledge to the task exhibited higher confidence than judges who could do so. But, in a sense, the Dependent condition acts as a control because it shows the levels of performance that could be expected from a judge with only external information sources. Data from this study suggest that such judges will be highly reactive to conflict between their advisors. Compared to Environments in which internal information sources can be accessed, the Dependent environment appears to produce the highest judge confidence with No Conflict, and lowest judge confidence with Conflict. In comparison, judges who can bring their own relevant knowledge to the decision task are less reactive, in terms of their final confidence, to advisor conflict.

Of course, sometimes it is desirable to alter the decision maker's uncertainty. If engendering high confidence in a judge is important, one should strive to approximate the situation in which the Dependent judges found themselves, that is, very low judge knowledge and complete agreement among advisors.

We had reason to believe that judges would become less confident when confronted with conflicting recommendations from their advisors. Sniezek and Kolzow (1994) tracked confidence assessments of individuals in groups and found individual confidence to decrease upon receipt of the discrepant judgments of other members. But in the present study, the effect of advisor disagreement on judge confidence was not to

decrease it relative to the individual baseline level, but to keep it from increasing over this level following advice. Whereas judge confidence tended to increase over the baseline level with agreement between advisors, it did not increase with disagreement. This is not to say that conflicting advice does not shake the confidence of a judge—only that any effect may be short-lived. In the Sniezek and Kolzow study, subjects began discussion after receiving the information and promptly regained—and soon exceeded—their initial levels of confidence.

We must point out that, despite our hypothesis, it is not reasonable to expect a one-to-one correspondence between disagreement and a reduction in confidence. Disagreement does not provide a pure manipulation of information processing in that it does not necessarily force judges to further evaluate the alternatives. As discussed in the section on judge accuracy, judges faced with disagreement might resort to heuristics that are independent of information about the alternatives. In such a case it would be possible for confidence to remain unchanged or to be inflated despite disagreement between advisors. The data for the Cued judges illustrate such a scenario. When faced with Conflict, they had the opportunity to evaluate the alternatives more completely. And because of the Cueing environment, they had not yet brought their own knowledge to bear on the task. Had they done so, further information processing could have reduced their confidence. Instead, Cued judges appeared to forego the opportunity to resolve the disagreement between their advisors by accessing their own knowledge base. They were not less confident than initial Independent judges but they were less accurate. Thus, it seems that, under Conflict, they would have been better off without advice.

Overconfidence

In the social setting of a Judge-Advisor team, judge overconfidence can be either subdued or magnified relative to unadvised individuals. In this study, the determinant of the direction of change is the interaction between Environment and Advisor Conflict. When compared to the Conflict condition, under No Conflict accuracy is improved more than confidence increases, thus, overconfidence decreases—unless one is in the Dependent condition. Our data under No Conflict reveal that Dependent judges were exceptionally overconfident (19.1). Although these were the judges who, because of the nature of the environment, could not bring their own knowledge to the task, they nevertheless expressed higher confidence and were less accurate than those judges (in the Cued and Independent conditions) who could do so. The implications of this extreme overconfidence for social decision making are

quite disturbing. But one should approach the results with caution because our data are limited in at least two ways. First, there was only one operationalization of the Dependent judge, and we have not yet established that such appalling overconfidence will be obtained with converging operationalizations. Because the Dependent condition is somewhat contrived (possibly forcing subjects to respond via some arbitrary heuristic) it remains to be seen if these results will be obtained with more representative designs. And, second, an ongoing relationship between advisors and a judge may involve sufficient communication and opportunities for learning so that judges are no longer completely dependent at the moment of choice.

Taken as a whole, the data on accuracy, confidence, and overconfidence have several implications for decision making in JASs. The tentative recommendation for optimizing performance is to make an independent choice prior to receiving advice. If it is not possible to choose independently, it probably won't hurt to take the advice of the more confident advisor—as long as one is aware that that person is not necessarily more correct. Another potentially useful suggestion that follows from these data is for the judge to beware of overreacting to advisor consensus by becoming more confident; the confidence expressed by advisors already tends to be too high relative to actual hit rates. In general, it is better to be in a situation in which one's advisors agree because one's final accuracy will be higher.

Social Influence

One objective of this study was to find out if confidence is an effective means of social influence. As used here effectiveness refers to the ability of advisors to get the judge to choose their answers—regardless of accuracy. The answer is clearly yes. Although the results of this study show that advisor confidence is not as powerful a persuader as is advisor consensus, it clearly has an important role in the judge-advisor relationship. When consensus exists, confidence plays a subservient role. In other words, when all advisors agree it does not appear to matter how confident they are—the sheer number (even when that number is only two) has overwhelming power regardless of Environment. But if there is disagreement among advisors, the effect of confidence becomes increasingly powerful. And, the strength of this power varies by condition in the expected direction such that Dependent > Cued > Independent. As the amount of personal information judges can bring (and do bring) to the decision task decreases, the power of an advisor to manipulate the judge's choice via confidence increases.

The Fig. 2 data on matching show how Independent

judges switch from their initial choice when that choice is opposed by both advisors. This "change of mind" on the part of judges can be traced to the normative influence (Kaplan & Miller, 1983; Stasser & Davis, 1981) under conditions of consensus between advisors. But what might cause a judge to shift answers when there is no consensus between advisors? The judge is then part of the majority, and shifting means succumbing to minority influence. While social psychological research has learned a lot about conformity to a majority, a complete understanding of social influence requires attention to occasions in which influence is exerted by a minority (Moscovici, 1985). A premise of the Judge-Advisor paradigm is that social influence is mediated by subjective uncertainty. The data concerning the shift of the Independent judge under disagreement suggest that a shift in final choice of the judge is more likely when the agreeing advisor expresses low confidence and the minority advisor expresses high confidence. Naturally this shift is further moderated by the confidence that judges have in their own choice.

In the Independent environment, the advisor with the higher confidence was able to change the decision of the (experimental) judge to agree with that advisor's position rather than that of the judge's (initially) and the advisor with the lower confidence on many occasions. In the Cued environment, advisors with high confidence were able to convince judges to agree with them and not the advisor with low confidence more often than in the Independent environment. In the Dependent condition the agreement with the advisor with high confidence was nearly complete—90.2% of the time. Considering that the Dependent judges could bring none of their own knowledge to the task and were left with a single clue (the relative confidence of their advisors), it is not startling to find that they nearly always agreed with the more confident advisor.

A more interesting situation concerns the difference in confidence utilization between the Cued and Independent judges. One explanation is that cueing may have limited the cognitive processing of the Cued judges and made them more susceptible to influence than were Independent judges. An alternative explanation is that confidence was more effective at influencing judge choice in the Cued condition because it did not have to compete with information about the judge's initial position as was the case in the Independent environment. There, advisor disagreement coupled with the judge's initial choice resulted in a 2-1 split decision, leaving the judge with the option of attending to the majority or to the confidence of the disagreeing advisor.

Although confidence was less likely than consensus between advisors to change an Independent judge's mind, its power was clearly demonstrated. For exam-

ple, consider the case of the Independent judges who did not receive confidence assessments. If advisors disagreed, there was almost no chance to change the judge's mind. The judge's final answer almost always (93.4%) agreed with the advisor who agreed with the judge initially. This shows the strength of normative influence acting alone. But this was not the case when confidence assessments were present. As the initial confidence assessments of the judge and the confidence assessments of the agreeing and disagreeing advisors changed from case to case, the propensity of the judge to hold the initial position changed. The lower the initial confidence ratings of the judge and the agreeing advisor and the higher the confidence of the disagreeing advisor, the greater the likelihood that the opinion of the judge would be changed.

The enormous effects of confidence on the judge's final choices in the Conflict cells are found despite little change in accuracy. All accuracy levels in these cells ranged from 60% to 66%.

To understand the influence of confidence vs consensus in judge-advisor relationships, additional research must be conducted varying the number of advisors. This will allow the extent of advisor consensus to vary more widely, making future studies more representative. This should bring a better understanding of how confidence and cognitive conflict interact to change or produce choice alternatives and to reduce or increase uncertainty.

One goal of this research was to investigate the extent to which the level of advisor confidence can act to influence judges' choices and their confidence in the correctness of those choices. This study was successful in that it obtained clear evidence that, under conditions of disagreement, confidence can act as a form of social influence. And, depending on the situation in which the decision maker is operating, its influence can be quite powerful.

Confidence and Strategy

If confidence clearly affects the behavior of others, then confidence provides a means of influence. Those who communicate high confidence may get their way more often than those who are underconfident or even well calibrated. For members of organizations this may be vital. Consider the manager seeking a larger budget for a project or the person seeking a promotion or a raise. Accuracy (or even equity) may not matter. Under conditions of competition for limited resources, it may be that those expressing the most confidence in their respective judgments will get the resources they desire leaving the others to fight over what is left. An essential difference between this scenario and the procedures of this study is that the former involves a reward

structure that encourages competition while the latter had a structure that encouraged cooperation. Perhaps this explains why the exploration of causal effects on advisor confidence found none. Apparently the particular composition of variables in this study—those controlled and those manipulated—failed to produce any incentive for differential confidence on the part of advisors in any of the three Judge Choice conditions. Nevertheless, we encourage further endeavors to understand the antecedents of advisor confidence given its capacity to influence the choice of the judge. We anticipate that research on differential reward systems for advisors vs judges or for one advisor vs another may uncover the conditions that cause advisors to adjust their expressions of confidence and the utility of those adjustments to influence others.

Future Directions

The Judge-Advisor paradigm can represent many varieties of social decision making. Any one study, however, will necessarily be limited in the number of factors that can be examined. In this study we have chosen to control many variables for the opportunity to understand better the effects of a few. The variables that we omitted from our design and analyses may prove paramount in further applications of this research paradigm. Consider the complexity of relationships between judges and their advisors. They vary with respect to length, trust, communication, frequency, interaction mode, or formality. And, many personal factors that were held constant in this study (e.g., experience, self-esteem, age, status, gender) might be varied between the advisor and judge roles to provide better generalizability to actual decision situations. Manipulations of contextual variables such as reward size or reward probability may prove informative, as may systematic variations in the judge's opportunity to select advisors. Relationships are also dynamic; even interdependent decision makers in a laboratory setting can exhibit systematic changes in behavior over the course of an hour long experiment (e.g., Snizek, May, & Sawyer, 1990). And, as with all decision-making research, we can expect to find important effects due to the nature, amount, timing and source of feedback.

Conclusions

The data presented here, when considered with previous studies (Ronis and Yates, 1987; Snizek *et al.*, 1990), demonstrate that independent choice is qualitatively different from cued choice in a manner that suggests that it entails additional information processing in the initial evaluation of alternatives. It follows that Independent judges consider more information about the alternatives in their initial choice process (assum-

ing that they possess relevant knowledge) than do Cued judges.

Contrary to the findings related to independent choice, cognitive conflict did not appear to have a positive effect on information processing. Why? It may well have been a function of the task type. Other research on cognitive conflict used tasks that required estimation of quantities. This task required judges to make a choice. It may be that conflict reduces information processing in choice tasks. Or, it may be that this reduction occurs only when the number of alternatives is severely limited, as it was in this task. However, a more likely explanation is that, if our results are due to a reduction in information processing, this reduction results from the relatively minimal social nature of the task. In a task that allows for more complete communication, advisors would be able to provide the reasons for their choices. The communication of detailed reasons may, in fact, increase information processing as was expected here. Of course, it is quite possible that a combination of these factors is at work. Only further research can clarify the issue.

Despite the prevalence of social decision making and the evidence that it differs substantially from individual decision making, little experimental research has been done on the judgment and decision processes of multiperson systems. Given that important decisions often involve multiple parties, this paradigm holds promise for understanding and improving decision making practices.

This research has clearly shown that the choices and confidence—and therefore accuracy and overconfidence—or individuals who receive input from others differ from those of individuals who do not. Yet, it is also evident that some predictions based on theory and data from research on individual choice (Snizek *et al.*, 1990; Ronis & Yates, 1987) do hold for social decision making. Not surprisingly, cognitive conflict between advisors was related to accuracy and confidence, though not in the systematic manner expected. Further, there is evidence that confidence escalation through social interaction and the cueing effect can occur in this form of social decision making. We expect that judges in social settings are often cued prior to choice. For example, a decision problem along with its recommended solution may be introduced to a group leader during a meeting. Or, a coalition advocating a particular option may bring the decision problem to the judge. Another situation analogous to the Cued condition occurs when a manager delegates a decision problem to subordinates to obtain their advice without previously addressing the problem personally. In the hierarchical structure of an organization, where many JASs may be layered, the cumulative effects of cueing may be even stronger.

This project extends individual and group judgment, choice, and confidence research to the JAS. The differentiation of roles within the JAS makes it possible to represent the unequal legitimate status that exists in many social decision settings. And, the possibility of using explicit expression of confidence in choice allows new insights into the social influence common in decision making.

The hypotheses tested were based on established theoretical positions and varied in the degree to which they were supported. Nevertheless, generalizability of these findings to other judge-advisor settings awaits further inquiry. Judge-Advisor decision making under uncertainty is ubiquitous. Therefore, data concerning objective and subjective quality in JASs create a valuable addition to the individual and group decision-making literature.

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